

With the rapid rise of AI, there are unique risks associated with it.

When AI goes wrong, it usually takes the form of hallucinations (confidently inventing false information), algorithmic bias, or PR disasters caused by rogue, unmoderated chatbots. Because AI lacks human judgment and real-time comprehension, it requires continuous oversight. [1, 2, 3, 4, 5, 6]

Common Failure Types

- **Hallucinations & False Promises:** Generative models make things up—such as nonexistent court cases, bogus policies, or fake products. When these are deployed as corporate customer service or legal assistants, they can result in severe PR backlashes and liability issues. [1, 2, 3]
- **Algorithmic Bias:** AI learns patterns from historical data. If fed **biased or incomplete data, the system will actively perpetuate—and often amplify—**discrimination, such as rejecting certain demographics in job applications or misidentifying faces in security software. [1, 2, 3]
- **Prompt Hacking:** Users can exploit flaws in an AI's system by commanding it to "ignore previous instructions" or agree to all requests. This causes bots to unexpectedly swear at customers, mock their parent brand, or sell products for pennies. [1, 2]
- **Scale of Security:** AI multiplies whatever it interacts with. A basic vulnerability or poor-quality data source can be scaled into a catastrophic, company-wide data leak or system malfunction in mere seconds

Examples of failures.

<https://www.evidentlyai.com/blog/ai-failures-examples>

<https://www.cio.com/article/190888/5-famous-analytics-and-ai-disasters.html>

<https://www.thebci.org/news/when-ai-fails-everything-fails-differently-new-business-impact-analysis-bia.html>

<https://www.hks.harvard.edu/centers/carr/publications/how-ai-fails-us>

AI risk is one of the main reasons people and companies are cautious about AI adoption. At the same time, more companies are replacing people with agents.

We can start with failures due to biased or incomplete data (create a synthetic data set (1) drawn from a stationary distribution and then make it biased or (2) drawn from a non-stationary distribution and then remove some to make it incomplete) and analyze its impact on the decisions under some basic learning algorithms, if there is no human with

domain knowledge to intervene. We can compare the impact under non AI where data is used to predict the distribution and then optimize. This will give us some idea about the difference.

We can then come up with some remedies for quality insurance under AI. Emphasizing the importance of domain knowledge (knowing the business and what is going on), opensource (Wikipedia, digital democracy, collective intelligence platforms) approach to making it easier for people to intervene. What are the roles of frontline people? Accountability